

Manufacturing Industries

Social Science — Geography • Chapter 6 • Chapter Summary & Study Guide for Daksh

When raw cotton becomes a shirt, when iron ore becomes a car, when wheat becomes biscuits — that transformation is **manufacturing**. Industries are the backbone of a modern economy: they create jobs, earn foreign exchange, raise farm productivity, reduce poverty. This chapter explores India's manufacturing sector — from agro-based to high-tech, from public to private — and the issues of pollution and sustainability that come with growth.

Why are industries important?

- Modernise agriculture by supplying machines, fertilisers, seeds.
- Reduce dependence on agriculture by providing jobs in secondary & tertiary sectors.
- Eradicate poverty & unemployment by generating diverse employment.
- Earn foreign exchange via exports.
- Promote regional development if located in backward areas.

Classification of industries

We classify industries on several bases:

- **By raw material:** Agro-based (cotton, jute, sugar, edible oil, tea, coffee), Mineral-based (iron-steel, cement, aluminium, machine-tools), Forest-based (paper, furniture, lac), Marine-based (sea-food, fish-oil).
- **By product role:** Basic / Key (provide products for other industries — iron-steel, copper smelting), Consumer goods (directly for end users — sugar, soap, fan, toothpaste).
- **By capital invested:** Small-scale (low capital, lower employment per unit) vs Large-scale.
- **By ownership:** Public (Govt — SAIL, BHEL, ONGC), Private (Reliance, Tata, Bajaj), Joint sector (Govt + Private — Maruti Udyog), Cooperative (Amul, Sugar mills run by farmer cooperatives).
- **By bulk and weight of raw material:** Heavy (iron, steel) vs Light (electronics, garments).

Some major industries

- **Textile** (cotton + synthetic) — accounts for ~14% of industrial production and a quarter of total exports. Largest producer of cotton yarn but jute industry has declined since 1947 partition.
- **Sugar** — second-largest agro-based industry; India is one of the largest producers. Most mills in UP, Bihar, Maharashtra, Karnataka.
- **Iron-steel** — basic of basic industries. Plants: Jamshedpur (TISCO — first private), Bhilai, Rourkela, Bokaro, Durgapur, Vizag, Salem (SAIL). Located near coal/iron ore deposits and water sources.
- **Aluminium smelting** — second-most important metal industry; light, conducts well, doesn't rust. Plants: Korba, Renukoot, Hirakud.
- **Chemical industries** — fertilisers, drugs/pharmaceuticals, dyes, plastics; major hubs in Gujarat, Maharashtra.
- **Cement** — depends on bulky/heavy raw materials (limestone, gypsum, coal); Gujarat, Tamil Nadu, Andhra, Rajasthan.
- **Automobile** — Pune, Chennai, Gurgaon, Sanand, Jamshedpur. Driver of the secondary sector since 1991 reforms.
- **IT & electronics** — software services, BPO, hardware. Bengaluru, Hyderabad, Pune, NCR, Chennai. India is a global IT/services hub.

Industrial pollution and sustainable development

Industries contribute to four kinds of pollution:

- **Air pollution** — particulate matter, SO₂, NO_x from chimneys (steel, cement, thermal power).
- **Water pollution** — untreated effluents into rivers (paper, chemicals, dyes).

- **Land pollution** — solid waste, fly ash, industrial dust.
- **Noise pollution** — heavy machinery, hydraulics.

Control measures: minimising water use, reusing/recycling water, treating effluents in three stages (primary mechanical, secondary biological, tertiary chemical), installing scrubbers, electrostatic precipitators and bag-house filters, locating polluting industries away from cities, implementing the **Polluter Pays Principle**.

NTPC and the way forward

NTPC (National Thermal Power Corporation) is given as an example of a major power producer that has adopted environmental policy: minimising waste, reforestation, ash-pond management, and adopting newer cleaner technology. Sustainability requires industry, government and citizens to work together.

★ MUST-REMEMBER POINTS

- Three sectors: Primary, Secondary (manufacturing), Tertiary.
- Classification by raw material (agro/mineral/forest/marine), product role (basic vs consumer), capital, ownership.
- Iron-steel plants: Jamshedpur (TISCO, private), Bhilai, Rourkela, Bokaro, Durgapur (SAIL).
- Cotton textile = largest organised industry, big exports earner.
- Aluminium smelting plants: Korba, Renukoot, Hirakud.
- Industrial pollution: air, water, land, noise. Polluter Pays Principle.
- Sustainable development requires policy + technology (e.g. NTPC example).

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